A RELATIVE GLOBAL COMPLETE INTERSECTION IS DEFINED OVER A FINITELY GENERATED SUBRING OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted statement refers to the definition of a relative global complete intersection; it is recorded as a navigation stub with kind definition, and the arguments using it stay inside the counted span.

Editorial note. The counted proof also refers to same-chapter statements by label, recorded the same way; no \cite appears, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Finitely many coefficients generate a subring. Used at the start of the counted proof (lines 36957-36959). Produces the finitely generated base over which the polynomials are already defined.
- Finiteness of the number of primes where a condition fails. Used in the counted proof at lines 36960-36980. Shows the defining conditions hold over the small base as well.

Source reference. Original definition 00SP: sources/stacks-algebra/pinned-original-source.tex, source lines 36723–36723. Navigation pointer for \ref{definition-relative-global-complete-intersection} used inside the counted span (line 36947); the argument that consumes it is retained in the span.

Source reference. Original lemma 00QH: sources/stacks-algebra/pinned-original-source.tex, source lines 31525–31525. Navigation pointer for \ref{lemma-dimension-fibres-bounded-open-upstairs} used inside the counted span (line 36967); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00P3: sources/stacks-algebra/pinned-original-source.tex, source lines 28507–28507. Navigation pointer for \ref{lemma-dimension-preserved-field-extension} used inside the counted span (line 36966); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ be a relative global complete intersection (Definition 00SP). There exist a finite type $\mathbf{Z}$-subalgebra $R_0 \subset R$ such that $f_i \in R_0[x_1, \ldots, x_n]$ and such that*

$$S_0 = R_0[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$$

is a relative global complete intersection.

Proof. Let $R_0 \subset R$ be the $\mathbf{Z}$-algebra of R generated by all the coefficients of the polynomials $f_1, \ldots, f_c$. Let $S_0 = R_0[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$. Clearly, $S = R \otimes_{R_0} S_0$. Pick a prime $\mathfrak{q} \subset S$ and denote $\mathfrak{p} \subset R$, $\mathfrak{q}_0 \subset S_0$, and $\mathfrak{p}_0 \subset R_0$ the primes it lies over. Because $\dim(S \otimes_R \kappa(\mathfrak{p})) = n - c$ we also have $\dim(S_0 \otimes_{R_0} \kappa(\mathfrak{p}_0)) = n - c$, see Lemma 00P3. By Lemma 00QH there exists a $g \in S_0$, $g \notin \mathfrak{q}_0$ such that all nonempty fibres of $R_0 \to (S_0)_g$ have dimension $\leq n - c$. As $\mathfrak{q}$ was arbitrary and $\mathrm{Spec}(S)$ quasi-compact, we can find finitely many $g_1, \ldots, g_m \in S_0$ such that (a) for $j = 1, \ldots, m$ the nonempty fibres of $R_0 \to (S_0)_{g_j}$ have dimension $\leq n - c$ and (b) the image of $\mathrm{Spec}(S) \to \mathrm{Spec}(S_0)$ is contained in $D(g_1) \cup \ldots \cup D(g_m)$. In other words, the images of $g_1, \ldots, g_m$ in $S = R \otimes_{R_0} S_0$ generate the unit ideal. After increasing R_0 we may assume that $g_1, \ldots, g_m$ generate the unit ideal in S_0. By (a) the nonempty fibres of $R_0 \to S_0$ all have dimension $\leq n - c$ and we conclude. $\square$